

Week 03

Title

Prime Palindrome Checking and Student Marks Calculation using Command Line Arguments

Objective

To develop Java programs to:

1. Check whether a given number is a **prime palindrome** using **Scanner**.
2. Calculate the **total marks and average marks** of five subjects using command-line arguments.

System Requirements

Hardware Requirements:

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Monitor

Software Requirements:

- JDK 8 or above
- VS Code / IntelliJ IDEA / Eclipse
- Command Prompt / PowerShell

Algorithm / Logic Description

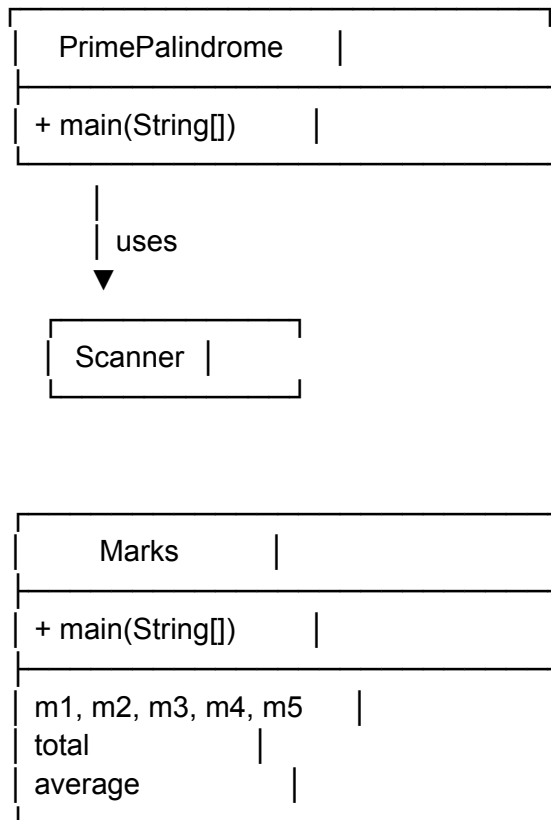
Part A: Prime Palindrome

1. Start the program.
2. Read a number using **Scanner**.
3. Check whether the number is prime by testing divisibility from **2** to **n-1**.
4. Reverse the number using the remainder operator **%**.
5. Compare the original number with its reversed number.
6. If the number is both prime and palindrome, display **Prime Palindrome**.
7. Otherwise, display **Not a Prime Palindrome**.
8. Stop.

Part B: Marks Calculation

1. Start the program.
2. Read five marks from command-line arguments.
3. Convert the arguments from `String` to `int` using `Integer.parseInt()`.
4. Calculate the total of five marks.
5. Calculate the average using `total / 5.0`.
6. Display the total and average.
7. Stop.

UML Class Diagram



Source Code

Save the file as **PrimePalindrome.java**.

```
import java.util.Scanner;
```

```
// Prime Palindrome Class
public class PrimePalindrome {
```

```
    // Main method
    public static void main(String[] args) {
```

```

// Creating Scanner object
Scanner sc = new Scanner(System.in);

int n, i, prime = 0;

System.out.println("Enter a number");
n = sc.nextInt();

// Checking whether the number is prime
for (i = 2; i < n - 1; i++) {
    if (n % i == 0) {
        prime = 1;
        break;
    }
}

// Finding reverse of the number
int temp = n;
int rev = 0;

while (temp > 0) {
    int dig = temp % 10;
    rev = rev * 10 + dig;
    temp = temp / 10;
}

// Checking prime and palindrome
if (prime == 0 && n == rev) {
    System.out.println("GIVEN NUMBER IS PRIME PALINDROME");
}
else {
    System.out.println("GIVEN NUMBER IS NOT PRIME PALINDROME NUMBER");
}

sc.close();
}

// Marks Class
class Marks {

    // Main method

```

```
public static void main(String[] args) {  
  
    // Reading marks from command-line arguments  
    int m1 = Integer.parseInt(args[0]);  
    int m2 = Integer.parseInt(args[1]);  
    int m3 = Integer.parseInt(args[2]);  
    int m4 = Integer.parseInt(args[3]);  
    int m5 = Integer.parseInt(args[4]);  
  
    // Calculating total marks  
    int total = m1 + m2 + m3 + m4 + m5;  
  
    // Calculating average  
    double average = total / 5.0;  
  
    System.out.println("Total = " + total);  
    System.out.println("Average = " + average);  
}  
}
```

Compilation and Execution Commands

Part A: Prime Palindrome

Compile:

```
javac PrimePalindrome.java
```

Run:

```
java PrimePalindrome
```

Sample Output

Enter a number

131

GIVEN NUMBER IS PRIME PALINDROME

Another example:

Enter a number

121

GIVEN NUMBER IS NOT PRIME PALINDROME NUMBER

Part B: Marks Calculation

The **Marks** class is inside the same **PrimePalindrome.java** file.

Compile:

```
javac PrimePalindrome.java
```

Run using five marks:

```
java Marks 80 75 90 85 70
```

Sample Output

Total = 400

Average = 80.0

```
PS C:\Users\navvee\OneDrive\Desktop\javaprograms\labprograms\Week 3> java primepalindrome
enter a number
131
GIVEN NUMBER IS PRIME PALINDROME
```

```
PS C:\Users\navvee\OneDrive\Desktop\javaprograms\labprograms\Week 3> java Marks 80
75 90 85 70
Total = 400
Average = 80.0
```